

Calvin Joyce

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Objective

Medical robotics R&D / New Product Development internship, **available Jan–Sep 2027**. Graduate researcher under **Prof. Michael C. Yip** (UCSD ARCLab); *Nature* co-author and 5× first author (ICRA, ISRR). Focusing on autonomy, controls, and learning for patient-contact robots, grounded in years of building the hardware myself, deploying on and improving real systems.

Education

University of California, San Diego · B.S. & M.S. Mechanical Engineering, GPA 3.8/4.0

B.S. 2024 · M.S. expected June 2027

Relevant coursework: Robotics, Haptics, Soft Robotics, Linear Control, Mechanical Design, Experimental Techniques

Skills

- **Medical Robotics & pHRI:** haptics & bilateral teleoperation, catheter/continuum robots, clinical safety thresholds, OptiTrack
- **Software & Control:** Python, C++, MATLAB, PyTorch, ROS 2; imitation learning (ACT), PPO, MPPI; NVIDIA Warp/Newton
- **Hardware:** SolidWorks + FEA, Fusion 360, DFMA, resin/FDM printing, laser cutting; design requirements & acceptance criteria, verification testing, test fixtures, capstan drives, motor drivers, sensors

Research & Work Experience

Advanced Robotics and Controls Lab (ARCLab), UC San Diego

June 2021 – Present

Graduate Student Researcher (funded); previously Full-Time Research Engineer, Undergraduate Researcher · Advisor: Prof. Michael C. Yip

La Jolla, CA

• PHABS (Portable Haptic Assisted Bimanual System)

Project lead; first author, ICRA 2027 submission

- * Attacking the force-blindness of video, simulation, and position-only teleoperation data, by capturing force-annotated bimanual demonstrations that anchor imitation-learning corpora.
- * Invented the first handheld bimanual device to render the internal force between two hands on a shared object, the signal that decides crushing, slipping, or tearing, by pairing capstan-driven pinch actuators with a gimbal-motor squeeze stage.
- * Closed the bilateral teleoperation loop on an Anvil OpenArm dual-arm robot, by mapping gravity-compensated joint effort through the manipulator Jacobian into per-hand pinch and signed inter-hand force channels.
- * Now training ACT policies on matched haptic vs. no-haptic demonstrations across three contact-rich bimanual tasks; verified force-rendering fidelity against a reference load cell.

• RIC (Robotic Infant Care) — first framework for direct robotic infant handling

Project lead; first author, ICRA 2027 submission

- * Defining safe robotic care for the NICU's most frequent physical interventions: bimanual infant pickup and CPAP nasal-mask repositioning, contact-rich tasks no robotic system has performed on a physical infant.
- * Formalized the safety bar, by deriving continuously measurable clinical thresholds with four NICU clinicians: cervical pitch and head acceleration for pickup; contact pressure, eye/mouth exclusion zones, and a PEEP-interruption time budget for CPAP.
- * Benchmarking direct-human, teleoperated, and autonomous (ACT) execution on a Unitree G1 humanoid, by building a manikin OptiTrack pipeline and a pressure-sensing molded CPAP mask, with every trial verified against the clinical thresholds.

• Humanoid Surgery (video) · in vivo feasibility of humanoid robots in laparoscopic surgery

Co-author, *Nature* 2026

- * Enabled humanoid robots to operate standard medical tools in dry-lab and in vivo porcine surgeries, by designing the tool-mount system (optimized mass/CoM, bench- and sim-validated rigidity) and running mechanical setup for every surgical session.
- * Coordinated a 10-person robot-clinician team through live surgeries, turning surgeon feedback into hardware revisions.

• Autonomous Catheter Navigation — continuum robotics; ARPA-H & DARPA project

Jul 2026 – Present

- * Building the learned low-level controller that drives a catheter tip to planner-issued waypoints, by training PPO policies in GPU physics with contact limits anchored to measured tissue perforation forces.
- * Moved training onto real patient anatomy, by building the pipeline from segmented mesh to trainable vessel geometry behind a one-command intake for any mesh.
- * Found and fixed a physics fault that had silently disabled the catheter's elasticity, by ruling out device, anatomy, drive, and stiffness with data before tracing it to our containment layer; also built a simulated C-arm rendering X-ray views from CT volumes.

• Mentorship: 16+ students across lab projects, through CAD/FEA design reviews, DFMA, fixture design, and test plans.

ArtemisAI Labs · Mechanical Engineer, La Jolla, CA

Jan 2026 – Present

- Made a NICU bedside camera mount clinically deployable, by writing its load, retention, and cleanability requirements (rough-handling loads benchmarked against the IEC 60601-1 instability test; no-disassembly hydrogen-peroxide wipe-down) and redesigning it against them with low-cost, manufacturable, and COTS components.
- Reached the bedside: the mount fields the cameras for NOVA, ArtemisAI's Phase II trial of computer-vision neurological monitoring of newborns at Mount Sinai; lining up assembly partners to kit and ship units to purchasing hospitals.

Triton Robotics, UC San Diego · Team President / Advisor

June 2020 – June 2023

- Scaled the team 20→80+ members and diversity 15%→35%+, by formalizing onboarding, DEI partnerships, and targeted recruiting.
- Ran a 17-lead organization with OKRs, roadmaps, and stand-ups, by founding business, marketing, and project-development teams; obtained national nonprofit status, \$3K in funding, and permanent lab space.

Selected Research Projects

• Haptic Shoulder (paper) — pHRI phantom; co-first author, ICRA 2025; provisional patent

Jan 2024 – June 2024

Invented a biomechanically accurate shoulder phantom for safe pHRI testing, by dedicating one motor-encoder module to each anatomical DOF to reproduce joint center, coupled kinematics, and configuration-dependent range of motion.

• ARCSnake V2 (paper) — amphibious screw-propelled snake robot, w/ NASA JPL; IROS 2026

June 2021 – Sep 2025

Increased screw output torque 40% by raising the torque ratio, aligning the belt drive, and cutting friction (shorter drive stack, relieved rubbing surfaces); engineered the mobile testbed that characterized screw performance across media (co-first author, ICRA 2023); designed NASU, the first Archimedes screw with reconfigurable angle of attack (first author, ISRR 2024).

Publications (Google Scholar)

• *Nature* 2026 (co-author): "In vivo feasibility study of humanoid robots in surgery"

• ICRA 2027 submission (first author): "PHABS (Portable Haptic Assisted Bimanual System) for Force-Annotated Bimanual Teleoperated Data"

• ICRA 2027 submission (first author): "How Safe Is Robotic Infant Handling? Benchmarking Learning from Demonstration Against Clinical Thresholds for Neonatal Pickup and CPAP Mask Repositioning"

• ICRA 2025 (co-first author): "Haptic Shoulder for Rendering Biomechanically Accurate Joint Limits for Human-Robot Physical Interactions"

• ISRR 2024 (first author): "NASU — Novel Actuating Screw Unit: Origami-Inspired Screw-Based Propulsion on Mobile Ground Robots"

• ICRA 2023 (co-first author): "Mobility Analysis of Screw-Based Locomotion and Propulsion in Various Media"

• Additional: "Humanoids in Hospitals" (arXiv 2025, co-2nd author); "LapSurgie" (ICRA 2026); "ARCSnake V2" (IROS 2026)